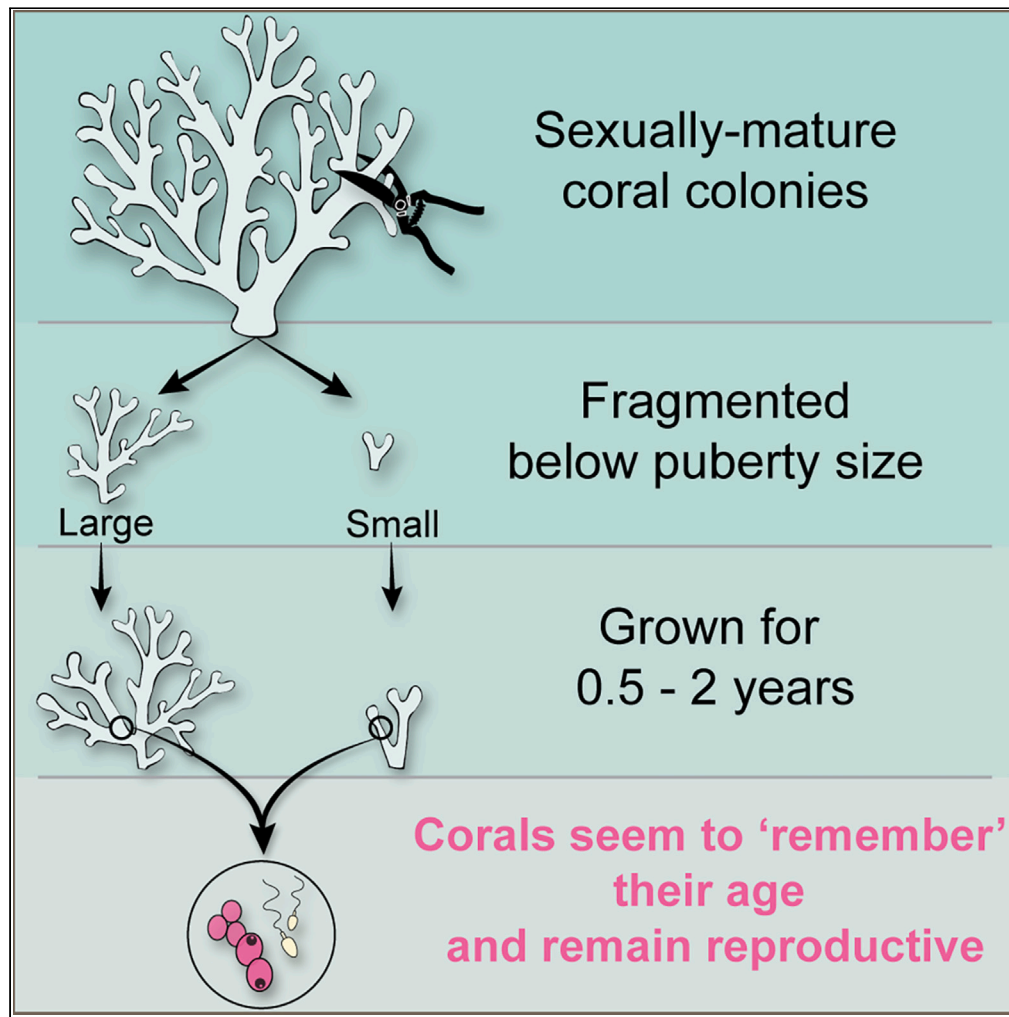


Article

Coming of age: Annual onset of coral reproduction is determined by age rather than size



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Highlights

Colonial corals maintain reproductive capacity even when greatly reduced in size

Corals seemingly "remember" their biological age

Corals, which are presumably non-aging, may experience aging after all

Rapuano et al., iScience 26, 106533
May 19, 2023 © 2023 The Authors.
<https://doi.org/10.1016/j.isci.2023.106533>

Article

Coming of age: Annual onset of coral reproduction is determined by age rather than size

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SUMMARY

Constraints on organisms possessing a unitary body plan appear almost absent from colonial organisms. Like unitary organisms, however, coral colonies seemingly delay reproduction until reaching a critical size. Elucidating ontogenetic processes, such as puberty and aging are complicated by corals' modular design, where partial mortality and fragmentation lead to distortions in colony size-age relationships. We explored these enigmatic relations and their influence on reproduction by fragmenting sexually mature colonies of five coral species into sizes below the known size at first reproduction, nurturing them for prolonged periods, and examining their reproductive capacity and trade-offs between growth rates and reproductive investment. Most fragments were reproductive regardless of their size, and growth rates hardly affected reproduction. Our findings suggest that once the ontogenetic milestone of puberty is reached, corals retain reproductive capacity irrespective of colony size, highlighting the key role that aging may have in colonial animals, which are commonly considered non-aging.

INTRODUCTION

Although modular, colonial organisms are common throughout many ecosystems and numerous phyla, many ecological and evolutionary theories are based on the biology and behavior of unitary organisms.^{1–3} Modular organisms grow through reiteration and cloning of basic individual units, producing colonies. The individual modules comprising a colony can typically survive and function independently, but are often integrated to form a cluster, coordinating different life functions and benefiting from shared resources, collective defense, and a size advantage, because the colony too functions as an individual unit.^{1,4–7} This modular structure facilitates resource acquisition across much of the colony surface, which may lead to different relationships between body size and metabolic rates (i.e., metabolic scaling) in colonial versus unitary organisms.^{8,9} The modular design also reduces some of the constraints on body size and growth that characterize unitary organisms, and may thus allow indeterminate growth.^{1,9,10} Colonial organisms generally feature efficient regenerative abilities,^{6,11,12} long life expectancy, and may not senesce like unitary organisms.^{2,13,14} Moreover, like plants with vegetative propagation, modular animal colonies may fragment into separate but genetically identical entities, creating non-linear relationships between size and age.^{15,16} The modular, clonal, and colonial nature of many plants and animals has generated numerous long-standing questions regarding the drivers and constraints of fundamental life functions such as growth, survival, sexual maturation, aging, and reproductive senescence.

Marine ecosystems are often dominated by colonial animals, such as corals, hydrozoans, ascidians, and bryozoans.^{4,17–19} Of particular ecological significance are scleractinian, hermatypic (i.e., reef-building) corals that form the complex reef structures upon which many other organisms depend. However, these ecosystems are severely threatened by local and global stressors, leading to coral mass-mortality events and hampering ecosystem recovery and resilience.^{20–23} There is a clear need, therefore, to elucidate fundamental biological and evolutionary processes involved in coral ontogeny and reproduction and their potential effects on coral-reef dynamics and persistence. Coral colonies are composed of polyps, the clonal, modular units of the colony. As these polyps multiply, the coral grows, with larger colonies having an advantage through shared resources and increased survivorship. Moreover, as the coral colony grows, its reproductive output grows as well through an increase in polyp number, fecundity, and the proportion of gravid polyps.^{16,24–31} Consequently, larger coral colonies may reproduce disproportionately more than smaller ones.^{25,31} However, corals in the wild below a species-specific size threshold do not appear to engage in sexual reproduction.^{16,26,27,31–35} Since their colonial nature enables partial mortality, fission, and

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<https://doi.org/10.1016/j.isci.2023.106533>



fusion,^{15,16,36} it is unclear whether coral sexual maturation and later investment in reproduction are regulated by colony size (i.e., limited size-dependent resources) or biological age (i.e., a developmental ontogenetic stage crucial for initiating gametogenesis).

In the early 1970s, Connell³⁴ noted a reproductive delay of some coral species until ~5–10 years old. He hypothesized that either (1) colony size is a key factor determining whether corals will invest in reproduction, or that (2) corals require time to reach maturity after which they remain sexually mature for many years. He proposed testing the first hypothesis by an experiment—fragmenting large, mature coral colonies into small pieces, followed by assessing their reproductive state. If a fragment smaller than the known size at first reproduction reproduces, the first hypothesis can be rejected. Answering this question is becoming increasingly important considering the current state of coral reefs and reductions in colony size and growth rates due to climate change and other anthropogenic disturbances.^{37–42} Nonetheless, five decades past Connell's³⁴ suggestion, very few studies have examined his hypotheses. The three studies that have, considered only four coral species with similar growth morphology—slow-growing, “massive” corals.^{43–45} While the earlier studies emphasized colony size as a probable determining factor for reproduction^{43,46} a more recent study concluded that size may drive sexual maturity, i.e. puberty, in one coral species but age in another.⁴⁵ Such species-specific differences coupled with the paucity of available information necessitate a study of a large variety of coral species, encompassing various life-history traits and morphologies, to reach a more generalized conclusion.

Here, we examined whether the annual onset of reproduction (i.e. gametogenesis, not to be confused with sexual maturity or puberty, which is the first onset of reproduction) in five species of colonial corals (listed in [Figures 1A–1E](#)) is a function of size or age, and how this affects reproductive investment. We experimentally fragmented large, sexually mature colonies of species with different reproductive strategies, life histories, and morphologies, into fragments of different sizes below the known sizes at first reproduction, that were ascertained during this study or obtained from the literature ([Table S1](#)). We then evaluated their reproductive state and growth rates after one to two annual gametogenic cycles and assessed possible trade-offs between growth and reproduction. Surprisingly, almost every fragment in every experiment was sexually reproductive ([Figures 1A–1E](#)). It appears that once corals reach the ontogenetic milestone of puberty, reducing colony size does not revert their reproductive state, which points to the importance of age in a colonial animal's life history.

RESULTS

Most coral fragments from all five species and experimental groups, were reproductive, including those that were substantially smaller than their species' known size at sexual maturity (e.g., single polyps of *Galaxea fascicularis* and ~1 cm fragments of *Acropora hyacinthus*). Of the large fragments, 95–100% contained gametes or planulae. Similarly, 71–100% of all small fragments were reproductive ([Figures 1A–1E](#)).

Reproductive performance of corals in different sizes

To ascertain the size at first reproduction of wild colonies from our focal species, we sampled small juvenile individuals during the reproductive season of the respective species, as well as large, presumably sexually mature colonies to verify the reproductive status of the wild colonies. While 75% of the large *Seriatopora hystrix* and 50% of the large *Pavona varians* were reproductive, none of the small *S. hystrix* (<4 cm) or *P. varians* (<5 cm) were reproductive ([Table S1](#)). We were not able to find small colonies of *A. hyacinthus* or *G. fascicularis* (see [STAR Methods](#) for further details), but based on the available literature, both species start reproducing at a size of ca. >5 cm. Data for *Turbinaria reniformis* were taken from our earlier study⁴⁷ performed at the same time and source population as in the current study, which showed that colonies smaller than 15 cm are not yet sexually mature.

Comparisons of the reproductive traits—proportion of gravid polyps, fecundity per polyp, and oocyte size, revealed only a few differences between the experimental groups and wild populations. Among all studied species, no significant differences were found in the proportion of gravid polyps between small and large fragments across experimental groups (differing in size and duration of growth) and between the experimental groups and colonies from the wild ([Table S2](#)). Similarly, no differences were found in fecundity ([Figures 2A and 2E](#)) or oocyte size ([Figures 3A and 3E](#)) in both *A. hyacinthus* and *T. reniformis*. Significant differences in fecundity between the experimental groups and wild colonies were only found in *S. hystrix*,

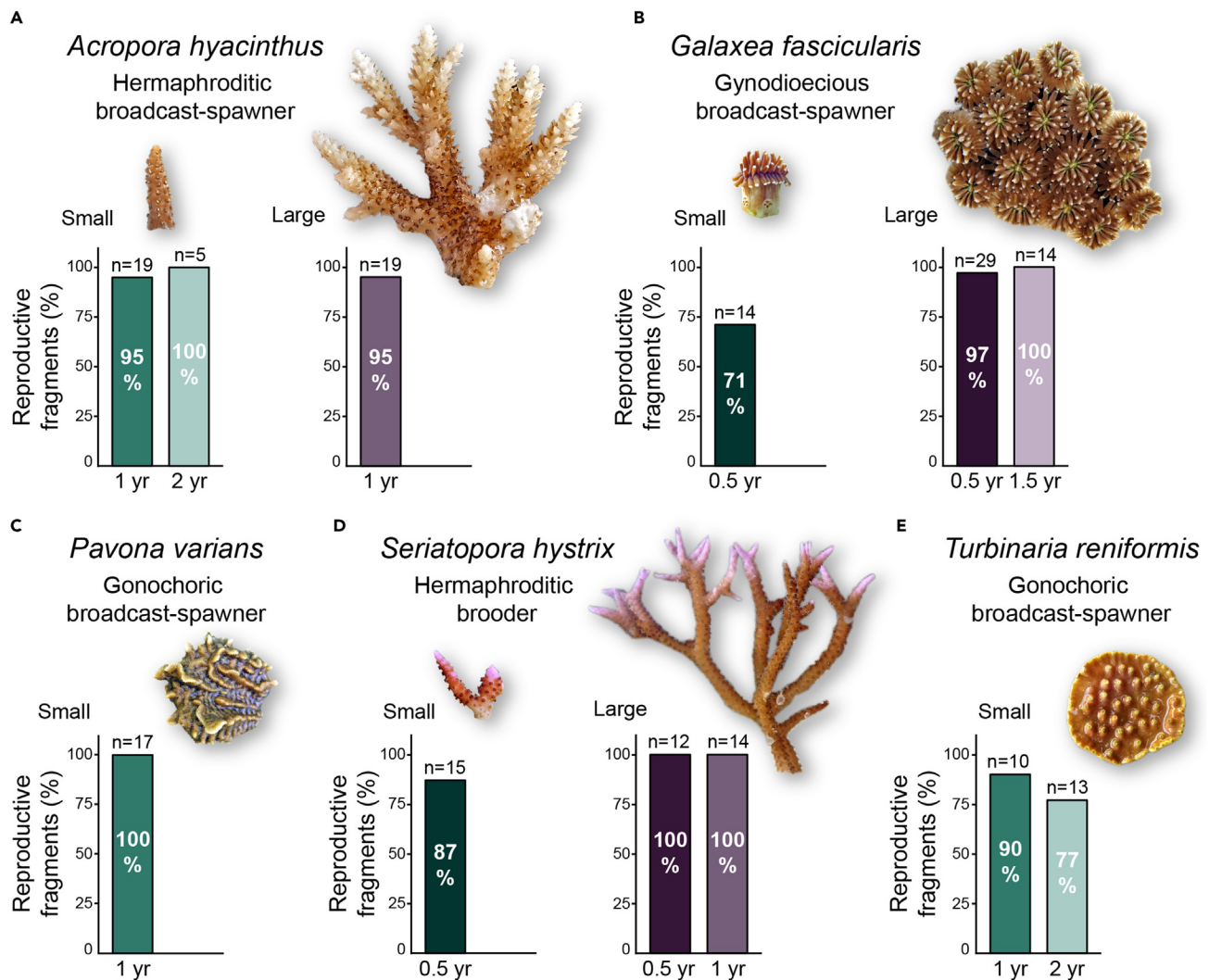


Figure 1. Percentages of reproductive fragments from small and large size-groups grown for varying durations, for each species

(A–E) n – number of fragments within the experimental group. Purple shades represent large fragments and green represent small fragments, with lighter shades indicating longer experiment durations.

with wild colonies being less fecund (permutation linear mixed model (PLMM), $p = 0.006$, Cohen's $d = 1.22$ for the small group; and $p = 0.024$, Cohen's $d = 0.79$ for the large group; Figure 2D). Similar comparisons between different-sized fragments and between the experimental group and wild colonies for *P. varians* are not available (see STAR Methods). The only species in which a significant difference in fecundity between experimental groups was found was *G. fascicularis*, with small fragments grown for 0.5 years being less fecund than large fragments grown for 1.5 years (pairwise comparisons, $p = 0.04$, Cohen's $d = 1.26$; Figure 2B).

Significant differences in oocyte sizes between experimental groups and wild colonies were only found in two species, *G. fascicularis* and *S. hystrix* (PLMM, $p = 0.002$ and $p = 0.003$, respectively; Figures 3B and 3D). Wild colonies of *G. fascicularis* had larger oocytes than those of both large and small fragments grown for 0.5 years (pairwise comparisons, $p = 0.01$ for both, Cohen's $d = 1.26$ for small fragments and 1.75 for large fragments). Additionally, large fragments of *G. fascicularis* grown for 0.5 years in 2015–2016 had larger oocytes than both large and small fragments grown for a similar period in 2013 (pairwise comparisons, $p = 0.01$ and $p = 0.02$, respectively. Cohen's $d = 1.29$ and 0.93, respectively). In *S. hystrix*, propagule sizes (oocyte and planulae) of wild colonies were also larger compared to those from all experimental groups

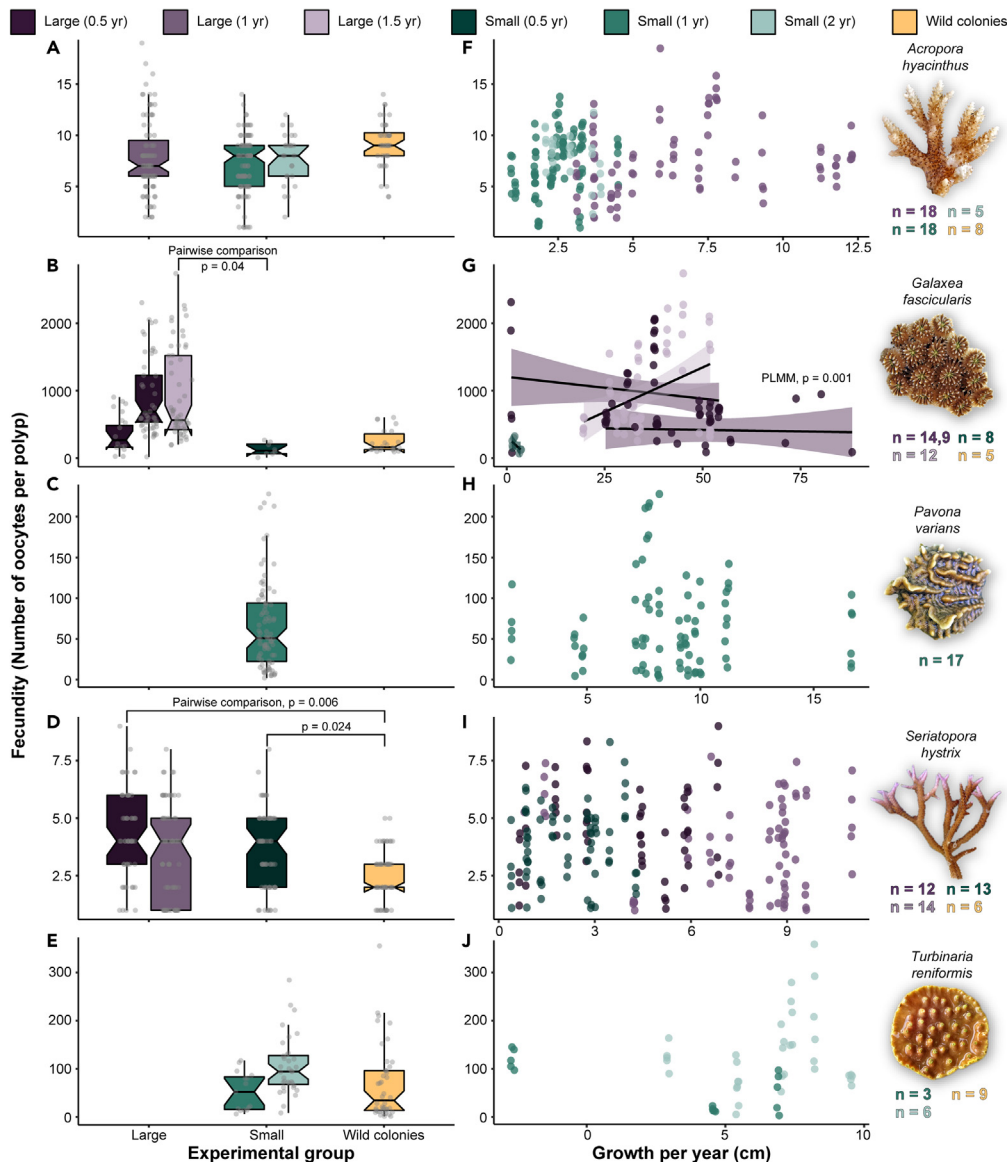


Figure 2. Effects of fragment size and growth rate on polyp fecundity

(A–E) Comparisons of fecundity between experimental groups (grouped by fragment size, i.e., “small” and “large”, and duration of growth) and samples of wild colonies from each of the five focal species. Notched boxplots show the median (horizontal line), middle two quartiles (boxes), 95% confidence interval around the median (notches), and first and fourth quartiles (whiskers). Gray dots represent individual polyp fecundity measurements. p values indicate significance level of the pairwise comparisons (with Bonferroni’s correction).

(F–J) Fecundity as a function of growth per year (in cm for [F and I] and in cm^2 for [G, H, and J]). Colored dots represent individual polyp fecundity measurements. Where the effect of growth rate was significant ($p < 0.05$), regression lines are presented with the shaded area indicating a 95% confidence interval. PLMM – permutation linear mixed model. Note that measurements in (D and I) are comprised both oocytes and planulae, referred to as “propagules” above. Sample sizes of each group are given below each species illustration.

(pairwise comparisons, $p = 0.006$ for all, Cohen’s $d = 2.08, 1.31$, and 2.43 , for large fragments grown for 0.5 years, large fragments grown for 1 year, and small fragments, respectively). Small *S. hystrix* fragments grown for 0.5 years did not show differences in propagule size compared with large fragments grown for 0.5 years but propagules were smaller than those of large fragments grown for a year (pairwise comparison, $p = 0.012$, Cohen’s $d = 0.87$).

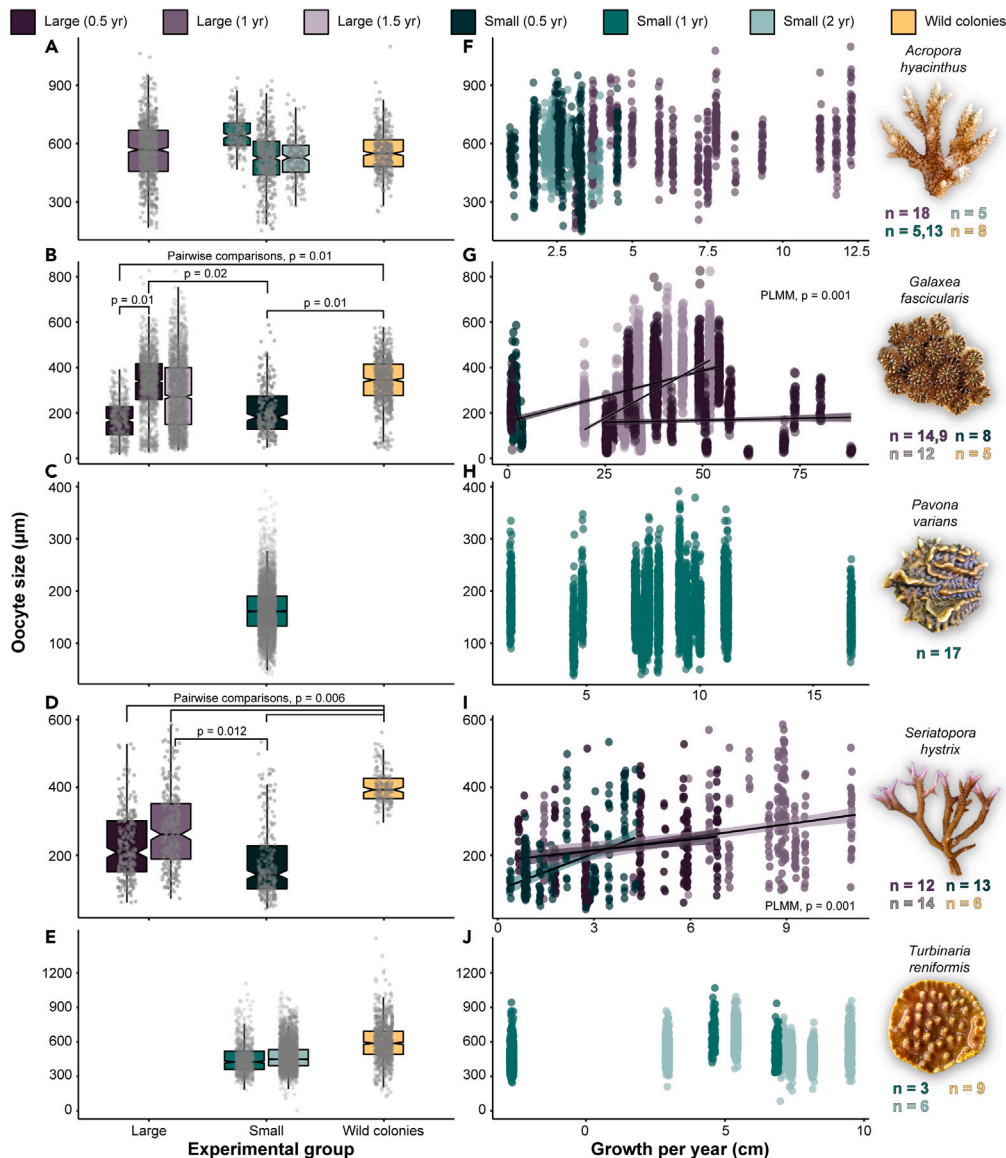


Figure 3. Effects of fragment size and growth rate on oocyte size

(A–E) Comparisons of oocyte size between experimental groups (grouped by fragment size, i.e., “small” and “large”, and duration of growth) and samples of wild colonies from each of the five focal species. Notched boxplots show the median (horizontal line), middle two quartiles (boxes), 95% confidence interval around the median (notches), and first and fourth quartiles (whiskers). Gray dots represent individual oocyte measurements. p values indicate significance level of the pairwise comparisons (with Bonferroni’s correction).

(F–J) Oocyte size as a function of growth per year (in cm for [F and I] and in cm² for [G, H, and J]). Colored dots represent individual oocyte measurements. Where the effect of growth rate was significant (p < 0.05), regression lines are presented, with the shaded areas indicating a 95% confidence interval. PLMM – permutation linear mixed model. Note that measurements in (D and I) are comprised both oocytes and planulae, referred to as “propagules” above. Sample sizes of each group are given below each species illustration.

Growth rates of corals at different sizes

Growth rates significantly differed between small and large size groups for *A. hyacinthus*, *G. fascicularis*, and *S. hystrix* (PLMM, p = 0.001 for all). Overall, small fragments demonstrated lower growth rates in comparison to larger fragments (Table 1). All small-size *A. hyacinthus* experimental groups grown for either 1 year (2015–2016 and 2017–2018) or for 2 years (2015–2017) demonstrated lower growth rates than those

Table 1. Summary of fragments' size and growth across the different species and experiments

Species	Size group	Duration (years)	n	Initial size	Final size	Growth per year	Relative growth (%)
<i>Acropora hyacinthus</i>	Large	1.0	19	Geometric mean diameter (cm)			
		1.0 (2017)	14	5.1 ± 1.2	11.5 ± 3.2	6.8 ± 2.8	135 ± 52
	Small	1.0 (2015)	5	1.7 ± 0.5	4.1 ± 1	2.5 ± 0.9	156 ± 61
		2.0	5	1.5 ± 0.2	3.7 ± 0.3	2.3 ± 0.3	152 ± 33
<i>Seriatopora hystrix</i>	Large	0.5	12	1.5 ± 0.2	7.2 ± 1.3	3 ± 0.7	198 ± 58
		1.0	14	6.6 ± 0.8	8.3 ± 1.6	3.8 ± 2.4	57 ± 32
	Small	0.5	15	4.6 ± 0.8	9.6 ± 1.6	7.6 ± 2.4	172 ± 161
		0.5	15	0.8 ± 0.1	1.8 ± 0.5	2.2 ± 1.1	274 ± 158
<i>Galaxea fascicularis</i>	Large	0.5 (2012)	15	Area (cm ²)			
		0.5 (2015)	14	12.4 ± 2.4	35.3 ± 10	53.3 ± 18.5	422 ± 98
		1.5	14	10.3 ± 6.7	42 ± 27.6	29.1 ± 17	314 ± 138
	Small	0.5	14	12 ± 4.4	57.7 ± 19.4	35.6 ± 9.2	334 ± 152
<i>Pavona varians</i>	Small	1.0	17	0.5 ± 0.12	1.5 ± 0.4	2.2 ± 0.8	469 ± 185
<i>Turbinaria reniformis</i>	Small	1.0	10	8.2 ± 3.1	15.2 ± 4.8	8.5 ± 3.1	113 ± 44
		2.0	13	12 ± 3.4	18.4 ± 6.5	9.2 ± 5.7	72 ± 41
		2.0	13	12.4 ± 3.5	22.1 ± 8.7	6.1 ± 3.9	46 ± 24

Average size and growth (±SD) of corals within the experimental groups, including sample size (n) and duration of the experiments. In *A. hyacinthus* and *G. fascicularis*, where two experimental groups grew for the same duration, the year of the experiment is indicated in parentheses.

in the large-size experimental group (pairwise comparisons, $p = 0.012$, $p = 0.012$, and $p = 0.024$, respectively). Small *G. fascicularis* fragments had lower growth rates than large fragments grown for 0.5 years in 2015–2016 and large fragments grown for 0.5 years in 2012–2013 (pairwise comparisons, $p = 0.024$ and $p = 0.012$, respectively). Large *G. fascicularis* fragments grown for 0.5 years in 2012–2013 demonstrated higher growth rates than the other two large-size experimental groups (pairwise comparisons, $p = 0.012$ for both). Small *S. hystrix* fragments and large fragments grown for 0.5 years showed lower growth rates compared to the large fragments that were grown for 1 year (pairwise comparisons, $p = 0.006$ for both).

By contrast, small fragments generally showed a higher relative growth rate on average (i.e., relative to their initial size), compared to large fragments (Table 1). While *A. hyacinthus* did not show a statistically significant difference in relative growth rates between experimental groups, the two other species, *G. fascicularis* and *S. hystrix*, did (PLMM, $p = 0.013$ and $p = 0.001$, respectively). In *S. hystrix*, relative growth rates were higher for the small fragments compared to large fragments grown for 0.5 years (pairwise comparisons, $p = 0.006$) and were also higher for large fragments grown for 1 year compared to those grown for 0.5 years (pairwise comparisons, $p = 0.006$).

Relationship between reproduction and growth

To explore possible trade-offs between coral investment in reproduction and in growth, we examined whether there is an effect of growth rate on reproductive traits among the experimental groups. Growth rate did not appear to have any significant effect on either fecundity (Figures 2F and 2H–2J) or oocyte size (Figures 3F, 3H, and 3J) in *A. hyacinthus*, *P. varians*, and *T. reniformis*. Only in *G. fascicularis*, there was a significant effect of growth rate on fecundity, which also differed significantly between experimental groups (PLMM, $p = 0.001$; Figure 2G), with a slight decrease in fecundity in most experimental groups and an increase in large fragments grown 1.5 years. Significant effects of growth rate on oocyte size were found in two species: *G. fascicularis* oocytes displayed an overall increase in size as growth increased, although there were differences between experimental groups (PLMM, $p = 0.001$; Figure 3G) with the small fragment group showing a decreasing trend in oocytes sizes. Likewise, *S. hystrix* propagules tended to increase in size as the growth rate increased (PLMM, $p = 0.001$; Figure 3I).

Age and size during puberty across Scleractinia

Our analysis, derived from published reports of puberty throughout the order Scleractinia (Figure 4, Table S1), revealed that scleractinian corals begin reproducing at a mean diameter (±SD) of 6.1 ± 4.1 cm (median = 5.0 cm,

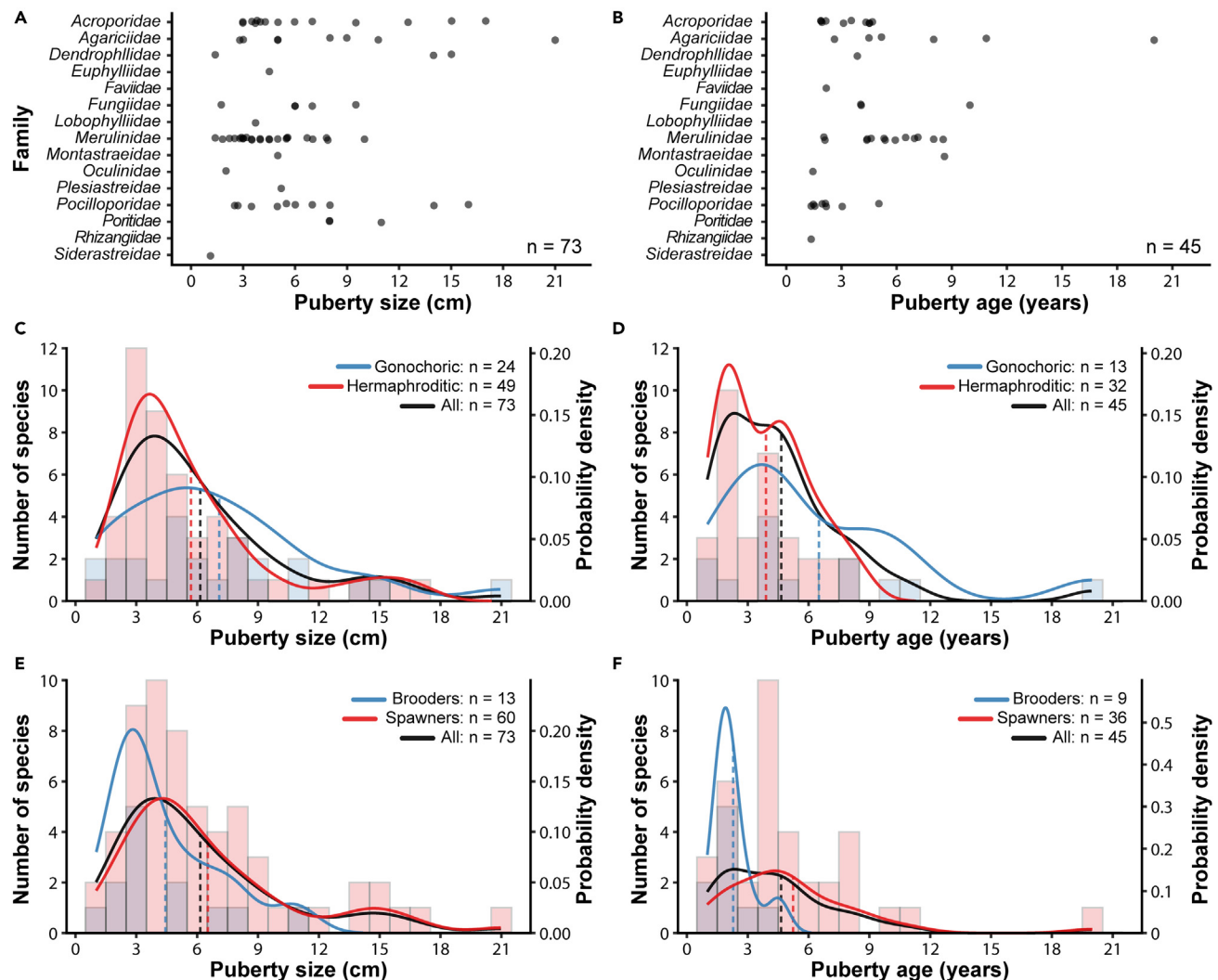


Figure 4. Coral size and age at puberty across Scleractinia

The extent of available data from literature pertaining to coral (A) size and (B) age at puberty, categorized by coral families. The distribution of gonochoric versus hermaphroditic coral species' puberty across (C) size and (D) age. The distribution of brooding versus broadcast-spawning coral species puberty across (E) size and (F) age. Histograms indicate the number of species, curves are the smooth probability density functions fitted to the data, and the dashed lines represent the means.

n = 73) and a mean age ($\pm$ SD) of 4.6 ± 3.4 years (median = 4.5 years, n = 45). We further explored patterns of age and size at sexual maturation among corals with different reproductive strategies. On average, hermaphroditic species start reproducing at smaller average sizes (5.7 ± 3.7 cm, median = 4.5 cm, n = 49) compared to gonochoric ones (7.1 ± 4.8 cm, median = 6 cm, n = 24). Similarly, hermaphroditic corals also start reproducing at a younger average age (3.9 ± 2 years, median = 4 years, n = 32) than gonochoric corals (6.5 ± 5.1 years, median = 4.5 years, n = 13). Brooding species appear to commence reproduction at smaller sizes (4.4 ± 2.7 cm, median = 3 cm, n = 13) compared to broadcast-spawners (6.5 ± 4.3 cm, median = 5.1 cm, n = 60). Similarly, brooders also begin reproducing at a younger average age (2.3 ± 0.9 years, median = 2 years, n = 9) compared to spawners (5.2 ± 3.5 years, median = 4.5 years, n = 36).

DISCUSSION

We set out to examine the extent to which sexual reproduction in colonial corals is driven by biological age versus colony size. We show that small corals that were fragmented from large, sexually mature colonies remain reproductive. This was true even after their size was reduced well below the natural size at first

reproduction. We found little effect of colony size or growth rate on most reproductive traits across all studied species. Overall, our findings indicate that the onset of coral sexual reproduction after having gone through puberty is not size-dependent, although both size and growth rate may influence reproductive investment (i.e., the mass of reproductive output). Our findings also suggest that colonial animals, such as corals, may “remember” their biological age and retain their reproductive ontogenetic stage after reaching sexual maturity.

It is generally believed that in young corals, characterized by small colony sizes, reproduction is delayed and most resources are diverted toward growth in order to overcome the high mortality rates common at smaller sizes, i.e., size-escape threshold.^{10,16,31,48,49} Indeed, large coral colonies can sustain breakage, partial mortality, and competition, better than small colonies, which may easily suffer whole-colony mortality.^{16,48} Consequently, it has been generally accepted that, for most coral species, reproductive onset and performance are governed by colony size,^{5,43,46,50} although this might be species-specific.^{36,45} Moreover, reproductive delay is rather poorly understood across metazoans, even in some of the most studied unitary organisms,^{51,52} let alone in colonial organisms. It is plausible that reproductive delay is a result of crucial developmental stages not yet having occurred, possibly connected to differentiation and development of stores of stem cells.^{50,53}

Furthermore, the individual modules constructing coral colonies and the basic unit of reproduction (i.e., polyps) require less time to reach a finite size compared to the time required for a colony to reach puberty size. Accordingly, if size is either the primary or only cause of reproductive delay at the colony level, individual polyps would require a mechanism of recognition of the overall colony size, thus triggering reproduction based on this recognition. Since in our study, almost all the coral fragments originating from sexually mature colonies retained their reproductive capacity even when fragmented below the known size of first reproduction, the onset of reproduction does not appear to be size-dependent; otherwise, we would expect most of these fragments to have invested mainly in growth and forgone reproduction. Consequently, we hypothesize that a delay in puberty may not necessarily be due to a size-driven energetic constraint but, rather, that a developmental threshold, based on an individual’s biological age, must be reached for a colony to become sexually mature. Yet, the drivers or constraints, such as colony size or age, on gametogenesis before reaching puberty may differ from those after puberty. Nonetheless, our results suggest that after passing the developmental milestone of puberty, corals retain the mechanistic “know-how” to reproduce.

As reductions in corals’ size and growth rates^{37–42} raise concerns regarding the ability of corals to reproduce and its demographic consequences, our findings offer an optimistic premise that even small but old enough corals will manage to reproduce. Additionally, the decline of many coral reefs worldwide is leading to rapid increases in restoration efforts.^{54–56} Yet, a critical knowledge gap in coral restoration ecology involves the role of coral age vs. size at sexual maturity⁵⁶ and therefore, the preferred size of fragments to be used, so that they will continue to reproduce sexually and enhance both genetic diversity and natural recruitment. Our study suggests that, given favorable environmental conditions, coral colonies can be fragmented in such a way that even small fragments remain reproductively viable.

Within the context of our study, it is important to note that the absence of gametes in similar experiments does not necessarily point to coral rejuvenation, nor is decreased reproductive investment the same as ceased reproduction. While reductions in reproduction can be attributed to various causes, among which are colony size reductions due to fragmentations and partial mortality,^{36,57–62} reproduction may also be halted altogether by acute stress. For example, acute stress in the form of starvation or predator attacks can trigger reproductive-inhibiting hormones in some vertebrates.^{63,64} Two earlier studies following Connell’s suggested fragmentation experiment³⁴ reported that sexually mature corals fragmented to below a size threshold did not reproduce.^{43,46} A later study reported that out of two studied species, only one reproduced following fragmentation.⁴⁵ In contrast to these early studies, here we show that across our five studied species, 169 out of 181 fragments (i.e., 93%) were sexually reproductive, with reproductive output that was generally similar to that of the respective species’ colonies in the wild. We speculate that the difference between the earlier studies and our study may arise from two main reasons: First, following the fragmentation of corals in the previous early studies, the corals were returned to the natural reef, potentially exposing them to a variety of natural disturbances and stressors, especially when being in a sensitive state of injury and regeneration. Second, those three earlier studies experimented with four species, all possessing a massive growth morphology, which is not naturally prone to fragmentation. Since mechanical

fragmentation of such species requires considerable force (i.e., using a hammer and chisel), this additional induced stress may have compounded the effect of reduced size in limiting reproduction. Moreover, since two of the aforementioned studies^{45,46} did not perform histological sectioning to observe the tissues at the cellular level, it cannot be ruled out that fragments reported as non-reproductive may have actually initiated gametogenesis, but gametes were too small to be recognized without a microscope. Since, in most of our focal species, we did not find any significant differences between reproductive traits in our experimental fragments and wild colonies that did not undergo experimentation, we believe that they were not unduly stressed by the experimental conditions, and thus represent the fundamental biology of unstressed corals. Another possible explanation for the difference between our results and past studies may be related to the lower seasonal water temperature variability in the Gulf of Aqaba and Eilat compared to the temperature fluctuations where some of the earlier studies were conducted. The lower water temperature fluctuation in our study region may be comparatively more comfortable for corals to invest more in reproduction and less in survival and regeneration following fragmentation injuries.

Like all other organisms, corals manage a presumably limited resource budget (commonly discussed in the form of energy, lipids, metabolites, or nutrients) by allocating resources toward the changing demands of biological functions, such as growth, maintenance, and reproduction.^{3,28,65,66} While we found that a constraint in the form of colony size reduction in sexually mature corals might not be enough to arrest reproduction altogether, it does appear that some species may trade-off between resources allocated to reproduction and other basic life functions. Although corals are efficient nutrient recyclers, there is evidence that they may be nutrient limited depending on environmental conditions and species-specific trophic ecology.⁶⁷ However, it is uncertain what effect a reduction in size would have on coral nutrient acquisition. In fact, small colonies may be more efficient at mass transfer with higher rates of diffusion than large colonies,³⁷ which does not seem to support size-dependent nutrient limitation as an explanation for reproductive delay. Moreover, a reduction in fecundity in damaged or fragmented corals has usually been explained as the need to direct the limited available energy toward regeneration and growth.^{43,57–60,65} However, when not under chronic stress, some evidence points to shallow-water corals not being energetically constrained, even when undergoing regeneration.^{53,68–70} Rather, reproduction, regeneration, and growth may be competing for a different crucial resource—stem or interstitial cells.⁵³

Although coral growth rates and relative growth (i.e., growth relative to initial size) were both measured, we tested the effects of growth rates on reproductive traits rather than relative growth, since proportional increases in coral growth rate decline with colony size and may therefore be mistakenly conceived as higher growth rates in small colonies compared to large colonies.⁷¹ We found species-specific associations of reproductive effort with growth rates: *G. fascicularis* and *S. hystrix* demonstrated an overall increase in reproductive output (either fecundity or oocyte size) as growth rate increased while in the three other species growth rates did not seem to affect reproductive investment. Similarly, we found very few effects of fragment size on reproductive traits: Only some of the large fragment groups of *G. fascicularis* and *S. hystrix* had larger or more propagules than small fragment groups, while for the rest of the studied species, no differences were found. It appears, therefore, that a reduction in size may have a minimal effect on reproductive traits in sexually mature corals. Interestingly, in the brooding coral *S. hystrix*, we did find an increase in the fecundity of both small and large fragment groups compared to the fecundity of wild colonies. Somewhat similar results were found in another brooding coral, *Pocillopora damicornis*, in Western Australia, where it was concluded that the overall output of fragmented corals has increased.⁶⁶ Together, such findings may be explained by a compensation response allowing a return to normal following the introduction of a stressor, whereupon increased or prolonged stress would then bring a decrease in fecundity.^{66,72}

Although it has been assumed that many processes (e.g., growth, reproduction, metabolism) in colonial animals should scale proportionally (i.e., isometrically) to body size,^{10,73} increasing evidence point to disproportionate scaling (i.e., allometric) being more common.^{9,31,74,75} Our reported species-specific differences in reproductive traits between groups of fragments of different sizes and between fragments and wild colonies may also reflect scaling differences between coral sizes. These interspecies variations in scaling may be a result of growth form and morphology variations between the studied species.^{8,9,37,75} Moreover, any level of resource allocation, whether in the form of energy stores, metabolites, or stem cells, is probably influenced by different life-history strategies and exposure to stress, so that under varying natural conditions, different species may still have different constraints involving colony size.^{36,45,65} For example, long-lived colonial cnidarians and bryozoans appear to have relatively large

size thresholds for puberty compared to short-lived colonial species, such as some hydrozoans and ascidians species.⁷⁶ Indeed, our extensive literature review of scleractinian corals' size and age at first reproduction (Table S1) shows that, on average, hermaphroditic species reach puberty at smaller ages and sizes than gonochoric species. Similarly, brooding species reach puberty at smaller ages and sizes than broadcast spawners. Furthermore, puberty size appears more variable among species than puberty age.

While our experiment utilizing sexually mature colonies provides a valuable first step toward resolving the drivers of ontogenetic processes and reproduction, the next obvious step would be a similar experiment including corals reared from larvae⁷⁷ or settled corals of known ages.⁷⁸ Using such corals, further disentangling the effects of age and size on reproduction can be achieved either by manipulating coral growth rates or by trimming back colonies, so that polyps potentially age while the colony remains small. Examining the effect of biological age (an indication of development and longevity) rather than simplistic, chronological age (measured in time)⁷⁹ can be potentially achieved by comparing DNA methylation profiles,⁸⁰ assessing mutation accumulation in somatic tissues,⁸¹ and developing biomarkers for senescence⁸² across different ontogenetic stages.

The debate surrounding the drivers of reproduction and puberty^{34,43,45} in colonial organisms emphasizes a much larger gap in our understanding of the control over ontogeny and aging, particularly regarding factors determining reproduction, indeterminate growth, and longevity potential. Our findings suggest that shrinking corals do not rejuvenate, but how could this tie to aging as well? The link between early ontogenetic development leading to puberty and aging lies in biological age. Biological age is rarely defined in the literature, although it is essentially an "indicator of the level of development, changes, or deterioration of a structure or function of an element of the organism, a functional system, or the organism as a whole."⁸³ Earlier theories of aging have postulated that senescence is inevitable.^{84,85} However, there is a growing realization that negligible or even negative senescence, characterized by a decline in mortality rate after sexual maturation and increased reproductive output with age, may be common throughout the Tree of Life.^{2,14,86,87} This may be particularly true for organisms with indeterminate growth such as modular or colonial organisms.^{2,14,86,88} Although historically regarded as non-aging,^{13,28,89} evidence from molecular aging markers, decline in growth, calcification, regeneration, or reproductive capacity among larger (and potentially older) corals, as well as their increased susceptibility to damage and stress, suggest that this may not be so.^{13,30,90–93} While ontogeny is sometimes defined as development of an organism from the fertilization of an egg until adulthood (i.e., sexual maturation), early discussions on ontogeny actually refer to it as the developmental processes throughout the entire lifespan of an individual up until death.⁹⁴ As such, ontogeny should be seen as a forward trajectory driven by biological age, where early development toward puberty and aging later in life are not isolated processes but part of a continuum.^{83,95–97} Indeed, some theories on the origins of aging posit that certain mechanisms controlling puberty and development may be connected or similar to processes mediating aging^{95,97}; and, accordingly, some animals were found to experience longer lives when puberty was delayed.⁹⁸ If colonial animals such as corals do age, this may be a process that already starts early on in life.⁹⁶

Here, we have shown that, for coral colonies, crossing the ontogenetic milestone of puberty is permanent and may be a greater determinate than size regarding the potential to engage in sexual reproduction. Our results highlight the importance of biological age in regulating coral puberty and further hint at the possibility that aging, of which so little is known in long-living colonial animals, may play a key role in their biology and ecology.

Limitations of the study

Some of the earlier studies^{43,45,46} following Connell's suggested experiment³⁴ have concluded that size or age may drive puberty. However, we would like to point out that in those studies, as well as in our current study, fragmentation of post-pubescent corals cannot point to the trigger for puberty/sexual maturity being size or age, since drivers of onset of reproduction may be different before and after puberty. Therefore, this study can point only to whether the annual onset of reproduction is primarily limited by size in post-pubescent colonies.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.isci.2023.106533>.

ACKNOWLEDGMENTS

We thank the Underwater Observatory Marine Park at Eilat for providing the infrastructure and support for our experiments. We also thank the Interuniversity Institute for Marine Sciences in Eilat (IUI) for providing further logistical support, and the Loya and Bronstein lab mates for their counsel and support in the field. This research was supported by grants from the Israel Science Foundation ISF-NRF (The National Research Foundation of Singapore) joint research program grant no. 2654/17 to Y.L., PADI Foundation and the Interuniversity Institute of Marine Sciences in Eilat to H.R., Tel Aviv University start up package to O.B., and grants from the Israeli Taxonomy Initiative, Interuniversity Institute for Marine Sciences at Eilat, Rieger Foundation, and PADI foundation to T.S.

AUTHOR CONTRIBUTIONS

Y.L. initiated the study. H.R., T.S., O.B., and Y.L. conceived and designed the study. H.R. and T.S. performed the field work, and T.S. carried out the experiments. H.R., L.R., and T.S. performed the laboratory work. H.R. and T.S. analyzed the data and produced the figures. Y.L. and O.B. supervised the field work, lab-work and writing. The first draft of the paper was written by H.R. with input from T.S., and all authors edited and approved the final version.

DECLARATION OF INTERESTS

The authors declare no competing interests.

Received: October 10, 2022

Revised: January 19, 2023

Accepted: March 24, 2023

Published: March 31, 2023

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Biological samples		
<i>Acropora hyacinthus</i>	Eilat coral Reef (2-12 m depth), Gulf of Aqaba/ Eilat, Northern Red Sea (29°30'05.9"N 34°55'01.7"E)	Nature and Park Authority, Israel, permit no.: 2012/38798, 2014/40249, 2015/40882, 2020/42527
<i>Galaxea fascicularis</i>	Eilat coral Reef (2-12 m depth), Gulf of Aqaba/ Eilat, Northern Red Sea (29°30'05.9"N 34°55'01.7"E)	Nature and Park Authority, Israel, permit no.: 2012/38798, 2014/40249, 2015/40882, 2020/42527
<i>Pavona varians</i>	Eilat coral Reef (2-12 m depth), Gulf of Aqaba/ Eilat, Northern Red Sea (29°30'05.9"N 34°55'01.7"E)	Nature and Park Authority, Israel, permit no.: 2012/38798, 2014/40249, 2015/40882, 2020/42527
<i>Seriatopora hystrix</i>	Eilat coral Reef (2-12 m depth), Gulf of Aqaba/ Eilat, Northern Red Sea (29°30'05.9"N 34°55'01.7"E)	Nature and Park Authority, Israel, permit no.: 2012/38798, 2014/40249, 2015/40882, 2020/42527
<i>Turbinaria reniformis</i>	Eilat coral Reef (2-12 m depth), Gulf of Aqaba/ Eilat, Northern Red Sea (29°30'05.9"N 34°55'01.7"E)	Nature and Park Authority, Israel, permit no.: 2012/38798, 2014/40249, 2015/40882, 2020/42527
Chemicals, peptides, and recombinant proteins		
Formic acid 98-100%	Merck	CAS no. 64186
Tri-Sodium citrate dihydrate	Merck	CAS no. 6132043
Formaldehyde 35% AR	Bio-Lab	Cat no. 067505
Software and algorithms		
NIS Elements D 3.2 software	Nikon	Version 2.2
R (A language and environment for statistical computing)	R Core Team	Version 4.04
Photoshop CS6	Adobe	Version 13.0x64
Other		
Eosin 1% Aqueous solution	Kaltek SRL	Product code: 1119
Mayer Hematoxylin solution	Kaltek SRL	Product code: 1100
Stereo Microscope	Nikon	Model SMZ1500
Digital Microscope	Nikon	Model Eclipse 90i
Microscopy camera (mounted on microscope)	Nikon	Model DS-Ri1

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be directed to and will be fulfilled by the lead contact Hanna Rapuano (nosponge@yahoo.com).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- All data generated and analyzed are available in the main text, [STAR Methods](#) or in the [supplemental information](#).

- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Studied species

We performed this study on five common coral species, widely distributed throughout the Indo-Pacific. These five species belong to five different families and represent a range of both reproductive strategies (e.g., broadcast-spawning and brooding, hermaphroditism and gonochorism) and growth morphologies (e.g., encrusting, branching, plating): (1) *Acropora hyacinthus* (Dana, 1846) (family: *Acroporidae*) is a branching tabular coral, reproducing as a hermaphroditic, broadcast-spawner.^{33,99} (2) *Galaxea fascicularis* (Linnaeus, 1767) (family: *Euphylliidae*) is a sub-massive to massive coral, reproducing by broadcast-spawning but exhibiting a unique sexual system of gynodioecy or pseudo-gynodioecy.^{100,101} (3) *Pavona varians* Verrill, 1864 (family: *Agariciidae*) is an encrusting to laminar or sub-massive coral, reproducing as a gonochoric or sequential hermaphroditic broadcast-spawner.^{102,103} (4) *Seriatopora hystrix* Dana, 1846 (family: *Pocilloporidae*) is a delicate branching coral, reproducing as a hermaphroditic brooder.^{104,105} (5) *Turbinaria reniformis* Bernard, 1896 (family: *Dendrophylliidae*) grows as tiered whorls comprising plates and unifacial laminae and reproduces as a gonochoric broadcast-spawner.^{24,47}

METHOD DETAILS

Experimental design

To assess the effects of age versus size on coral reproductive performance and growth rates, we carried out a series of experimental fragmentations between 2012 and 2018. Each year we experimented with a different set of species that were fragmented into two size classes prior to the onset of their gametogenic cycles. Fragment sizes were smaller than the species' known size at first reproduction, as determined in the current study or reported in the literature (Table S1). In each experiment, fragments were carefully generated by using sharp pruning shears. Fragments were generated from 3–10 large, sexually mature colonies collected in the Gulf of Aqaba and Eilat, northern Red Sea (29°30'05.9"N 34°55'01.7"E). These fragments were carefully nurtured *ex-situ* in an open flow-through water system, for periods spanning 0.5 to 2 years and encompassing at least one gametogenic cycle. Final sampling of the fragments took place just prior to the peak of their reproductive season, based on earlier studies.^{22,47,102,104} To assess possible inter-annual variations and to validate the reproducibility of our results, we repeated some of the experiments for four of the five focal species in different years.

In December 2012, we collected three colonies of *G. fascicularis* and four colonies of *S. hystrix* at 6–12 m depth. From the *G. fascicularis* colonies we created 14 small fragments, featuring only a single polyp (~1 cm in diameter), and 15 large fragments, featuring 10–20 polyps (~3–5 cm in diameter). From the *S. hystrix* colonies, we created 15 small fragments (~1 cm length) and 12 larger fragments (~6–8 cm length). All fragments were grown for half a year and sampled on June 17, 2013. In 2015, we repeated the experiments with the abovementioned two species, but only created the large size-group fragments from four colonies of each species. In September 2015, *S. hystrix* colonies were fragmented into 14 fragments and grown for almost one year until sampled on June 12, 2016. In November 2015, *G. fascicularis* colonies were fragmented into 28 fragments and half of them were sampled half a year later, on June 13, 2016, while the remaining fragments were sampled a year and a half later, on June 15, 2017. In June 2015, we created ten small fragments, ~1–2 cm in length, and four larger fragments of ~5–8 cm length, from ten different colonies of *A. hyacinthus* collected at a depth of 0–2 m. The four large fragments of *A. hyacinthus* and five of the small fragments were sampled a year following fragmentation (13 June 2016), while the other five small fragments were sampled two years later, on 7 June 2017. In 2017 we repeated the experiments with *A. hyacinthus* while increasing the number of replicates. In June 2017, we created 14 small and 15 large fragments from five colonies, and sampled these fragments a year later, on May 21, 2018. Experiments with *P. varians* and *T. reniformis* were based only on small fragments (~2–5 cm in diameter) from four and ten colonies respectively, collected at a depth of 4–8 m. We created 23 fragments of *T. reniformis* as follows: in September 2015, we created 13 fragments, of which eight were sampled almost a year later, on June 13, 2016, and the other five following two years, on June 15, 2017. In 2016, we created nine additional fragments of *T. reniformis* and sampled three of them following one year on June 15, 2017, and six following two years, on June 29, 2018. An additional *T. reniformis* fragment was created in 2017 and sampled after

one year. Finally, in July 2017, we created 18 fragments of *P. varians* and sampled them following one year, on June 29, 2018. One large *A. hyacinthus* fragment and large *S. hystrix* fragment were missing growth data and were excluded from analyses related to growth. One *P. varians* fragment underwent substantial partial mortality and was similarly excluded from all analyses.

Measurements of coral reproduction and growth

To assess coral reproductive performance, fragment samples were dissected at the end of each experiment. Fragments were fixed in 4% formalin in seawater for 24 hours, rinsed in tap water and preserved in 70% ethanol. They were decalcified in 25% formic acid buffered with citrate (equal parts of 20% tri-sodium citrate in distilled water and 50% formic acid in distilled water)⁴⁷ and dissected under a stereo microscope (Nikon SMZ1500). We measured four reproductive features: (1) oocyte size; (2) polyp fecundity; (3) proportion of gravid polyps within a fragment, and (4) proportion of gravid fragments within an experimental size-group. To determine oocyte sizes, we measured all oocytes in 5–10 haphazardly chosen polyps in each fragment. In samples with polyps that contained dozens to hundreds of oocytes, 30 randomly selected oocytes per polyp were measured. For fecundity measurements, we counted all oocytes in the polyps dissected. Since *S. hystrix* is a brooder, we pooled all propagules (i.e., both oocytes and planulae) for fecundity and size measurements, considering early-stage planulae and mature oocytes are indistinguishable in dissections. Since *G. fascicularis* polyps are especially difficult to dissect, oocytes were counted from only five mesenteries per polyp, averaged, and multiplied by the number of mesenteries in the polyp. All counts and size measurements were performed using NIS Elements D 3.2 software. We estimated the proportion of gravid polyps within a fragment in three of the five studied species. For both *A. hyacinthus* and *S. hystrix*, estimations were based on 20 randomly selected polyps that were scored for presence or absence of gametes. In *T. reniformis*, presence or absence of gametes was estimated in all polyps (maximum of 47 polyps) within the sample. Similarly, for estimating the proportion of gravid fragments in each experimental group, entire fragments were scored for presence or absence of male or female gametes identified in dissection. In some cases where gametes were too small to be clearly identified during the dissections, histological slides were prepared by dehydrating and embedding a ~1 cm piece of tissue, cross-sectioning it at 7 μm, mounting on slides and staining with Mayer's hematoxylin and Putt's eosin (H&E),⁴⁷ and presence or absence of gametes was verified after examining the slides under a microscope (Nikon Eclipse 90i).

In parallel to the reproductive estimations, we assessed the growth rate of the coral fragments by measuring their size at the beginning and end of the experiments. We calculated annual growth rates using the following equation:

$$(Final\ size - Initial\ size) \times \frac{365}{Total\ growth\ days} \quad (Equation\ 1)$$

Relative growth rates were calculated by dividing the annual growth rates by the fragment size at the beginning of the experiment. Fragments of the branching corals, *A. hyacinthus* and *S. hystrix*, were measured for perpendicular length, width, and height, and their size was estimated by calculating the geometric mean diameter:¹¹

$$(Length \times Width \times Height)^{1/3} \quad (Equation\ 2)$$

To determine the size of the three non-branching species, *G. fascicularis*, *P. varians*, and *T. reniformis*, each fragment was photographed, and its planar area measured using Adobe Photoshop CS6.

Comparisons with coral reproductive state in the wild

To compare the reproductive features measured in our experimentally fragmented corals with populations in the wild, we used recent published data on three of our experimental species from the same populations and years: Data for *G. fascicularis* and *S. hystrix* were taken from Shlesinger et al.,¹⁰⁶ and for *T. reniformis* from Rapuano et al.⁴⁷ and Shlesinger et al.¹⁰⁶ For comparisons of *A. hyacinthus*, we dissected samples from eight large (>25 cm) colonies from the Gulf of Aqaba and Eilat, on 28 May 2017, from a depth of 0.5–2 m.

To determine the size at first reproduction of each species, in 2020 we sampled the smallest colonies (~1–5 cm long) that could be confidently identified from the same local populations sampled for the fragmentation experiments (from depths of 5–15 m). Concomitantly, we sampled several large, presumably sexually mature colonies (i.e., >10 cm), to determine whether the wild population was reproductive. On

26 May 2020, four small *S. hystrix*, and six small plus seven large *P. varians* were sampled during their expected gametogenic cycle based on the literature.^{102,104} Later, on 12 August 2020, an additional five small and five large *S. hystrix*, and four small plus seven large *P. varians* were sampled. All samples were preserved and decalcified as described above and scored for the presence or absence of gametes. Since *P. varians* experience a relatively short gametogenic cycle compared to other species, our sampling was unknowingly too early in the season. Thus, we were able to confirm that the population was reproductive by histological methods, but gametes were too small for dissections used for comparisons between the experimental fragments and these wild colonies. Data on *T. reniformis* size at first reproduction were taken from Rapuano et al.⁴⁷ Despite the effort invested in searching for small, juvenile *G. fascicularis* and *A. hyacinthus* colonies, we were unable to find any. *Galaxea fascicularis* has already exhibited a prolonged absence of juveniles on the reefs of Eilat,²² while the absence of *A. hyacinthus* juveniles was probably a result of a major southern storm that hit the area in March 2020, prior to our sampling, and wiped out many corals.

Literature review: Coral size and age at puberty

We explored patterns relating to the onset of sexual maturation with size and age throughout the order Scleractinia. We reviewed the literature on coral reproduction and compiled available accounts of size and age at first reproduction (Table S1). This table provides additional information published after the original publication of Harrison and Wallace.²⁸ We then compared the distributions of species' size and age at sexual maturation between families, sexuality (gonochoric vs. hermaphroditic), and reproductive mode (spawners vs. brooders). Several sources had reported size of corals during sexual maturation according to the number of polyps, and these reports are included in Table S1, but excluded from the analyses. Similarly, several solitary or free-living Dendrophylliids, such as *Balanophyllia* spp., as well as one Caryophylliid species, were excluded from the analyses because even as adults they largely remain as small, solitary polyps (<~2 cm).

QUANTIFICATION AND STATISTICAL ANALYSIS

Coral fragments were grouped together according to their size (small or large) and duration of growth (0.5, 1, or 2 years; hereafter "experimental group"). Differences in oocyte size and fecundity between the experimental groups were assessed using linear mixed-effects models (LMMs) in which the random effects were individual polyps for fecundity comparisons and individual oocytes nested within polyp for oocyte size comparisons, with experimental size as the fixed effect. The proportion of gravid polyps within colonies, growth per year, and relative growth per year were likewise compared between the experimental groups using linear models. LMMs were also used to assess the effects of growth per year on oocyte size, fecundity, and gravid polyp proportions. In these models, growth and the experimental group served as the fixed effects, and individual polyps and individual oocyte nested within polyps were the random effects. When an experiment was repeated using the same size-group and duration, but in a different year, we assessed the differences between these experiments as described above. In the absence of differences, we pooled fragments belonging to similar durations and size-groups.

Full models, including interactions between growth per year and experimental group, were compared with models without interactions, and the best-fit models were selected using Akaike Information Criterion. The assumptions of parametric tests (e.g., homogeneity of variance, normality of residuals) were visually inspected, and when they were clearly violated, a permutation approach was taken. Statistical significance was considered when $p < 0.05$, and when significant differences or effects were found, the tests were followed by post-hoc analyses using Bonferroni's correction, and the effect size was estimated by calculating Cohen's d . All statistical analyses were performed using R v4.04¹⁰⁷ and all data underlying this study are available in Table S1–S3.